

Computer Science & IT

Discrete Mathematics



Set Theory & Algebra

Lecture No. 08



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Recap of Previous Lecture



Topic

Reflexive, symmetric and transitive closure



Topic

Equivalence relation



Topics to be Covered



Topic

Equivalence class



Topic

Partition of a set



Topic

Number of equivalence relation





Topic : Equivalence Class

Let "R" is an Equivalence relation on set A.

→ Equivalence class of an element $x \in A$ w.r.t. Equivalence relation R can be denoted by $[x]$, and it is defined as

$$[x] = \{ y \mid x \underset{\text{'''}}{R} y \}$$

i.e., $[x] = \{ y \mid (x, y) \in R \}$

Equivalence class of element 'x' is the set of all elements that relates with 'x' w.r.t relation 'R'

eg: Let $A = \{1, 2, 3, 4, 5\}$

and R is an Equivalence relation on set A .

$$R = \{(1,1), (2,2), (3,3), (4,4), (5,5), (1,2), (2,1), (4,5), (5,4)\}$$

Find Equivalence class of every element of set A
w.r.t. Equivalence relation R .

Soln. $[1] = \{1, 2\}$

Same
ie, $[1] = [2]$

$$[2] = \{2, 1\} = \{1, 2\}$$

$$[3] = \{3\}$$

$$[4] = \{4, 5\}$$

Same.
ie, $[4] = [5]$

$$[5] = \{5, 4\} = \{4, 5\}$$

Distinct equivalence
Classes of elements of
Set A w.r.t. relation R are

$$\{1, 2\}, \{3\}, \{4, 5\}$$

$\{\{1, 2\}, \{3\}, \{4, 5\}\}$ is a
Partition of set A

→ Note :- → ① Equivalence class of two distinct elements of the set may be equal

Note :- ② Set of all distinct Equivalence Classes of elements of set A will be a partition of set A with respect to given Equivalence relation

Gg: In the above example,
 $\{\{1, 2\}, \{3\}, \{4, 5\}\}$ is a partition of
Set $A = \{1, 2, 3, 4, 5\}$ with respect to
given Equivalence relation R



Topic : Partition of a set

→ Partition of a "set A " is the set of non-empty subsets of set A such that each element of set A appears in exactly one non-empty subset of that set.

→ Let A is any set and $A_1, A_2, A_3, \dots, A_k$ are non-empty subsets of set A .

$\{A_1, A_2, A_3, \dots, A_k\}$ is a partition of set A if and only if (i) $A_i \cap A_j = \emptyset, \forall i, j (i \neq j)$ and (ii) $\bigcup_{i=1}^k A_i = A$

eg: Let $A = \{1, 2, 3, 4, 5\}$
which of the following is/are partitions of set A.

✗ (a) $\{\{1, 2, 3\}, \{3, 4\}, \{5\}\}$

✗ (b) $\{\{1, 2\}, \{3\}, \{4\}\}$ { '5' is missing }

✗ (c) $\{\{1, 2, 3\}, \{4\}, \{5\}, \{\}$ } Empty subsets can not be a member of any partition

✓ (d) $\{\{1, 2\}, \{3\}, \{4\}, \{5\}\}$

✓ (e) $\{\{1, 2, 3, 4, 5\}\}$

✓ (f) $\{\{1, 2\}, \{3\}, \{4, 5\}\}$

→ Find the number of partitions of "set A" when

(1) $|A| = 0$

(2) $|A| = 1$

(3) $|A| = 2$

(4) $|A| = 3$

(5) $|A| = 4$

① Let $A = \{ \}$ i.e., $|A| = 0$

Partition of set $A = \{ \}$

it is the only
Partition of
Set $A = \{ \}$

There is no non-empty subset
of set A , i.e. There will be
no subset in the set.

Partition itself can be
an empty set; but
empty set can never be
an element of the set
w.r.t. partition

i.e., if $|A| = 0$, then number of partitions of set $A = 1$

② Let $A = \{a\}$ i.e., $|A| = 1$

Partition of set $A = \{\{a\}\}$

it is the only
Partition of
Set $A = \{a\}$

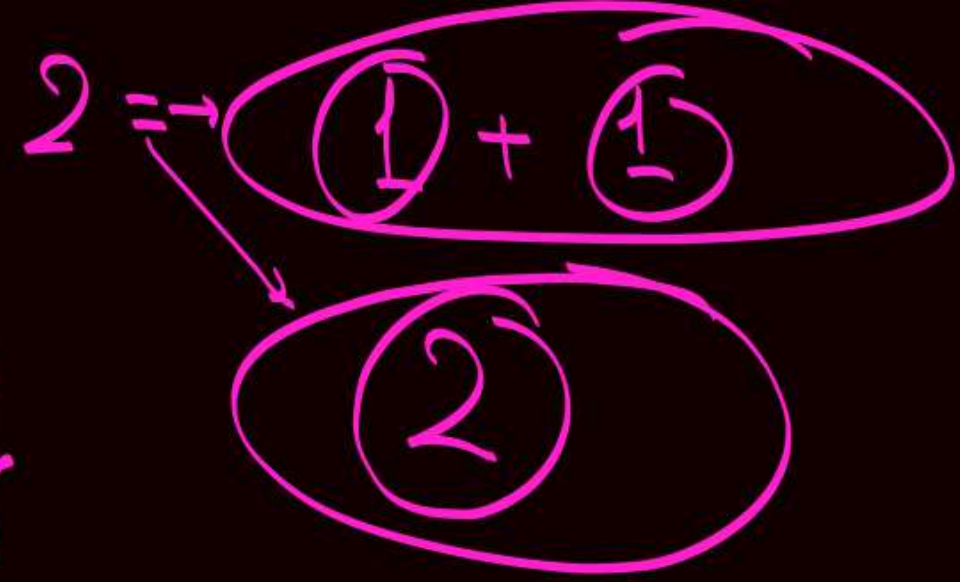
if Cardinality of the set is '1',
then there can be only one
non-empty subset and it
will be the set itself.

i.e., if $|A| = 1$, then number of partitions of set $A = 1$

③ Let $A = \{a, b\}$ i.e., $|A| = 2$
 $\{\{b\}, \{a\}\}$

Partition of set $A = \{\{a\}, \{b\}\}$ and $\{a, b\}$

These are the only two
Partitions possible w.r.t
Set $A = \{a, b\}$



i.e., if $|A| = 2$, then number of partitions of set $A = 2$

③ Let $A = \{a, b\}$ i.e., $|A| = 2$
 $\{\{b\}, \{a\}\}$

Partition of set $A = \{\{a\}, \{b\}\}$ and $\{\{a, b\}\}$

$|A|=2 \Rightarrow \underbrace{1 + 1}$
 it represent
 two subsets
 each of size = 1

$\Downarrow$
 $\{\{ _ \}, \{ _ \}\}$ (or)

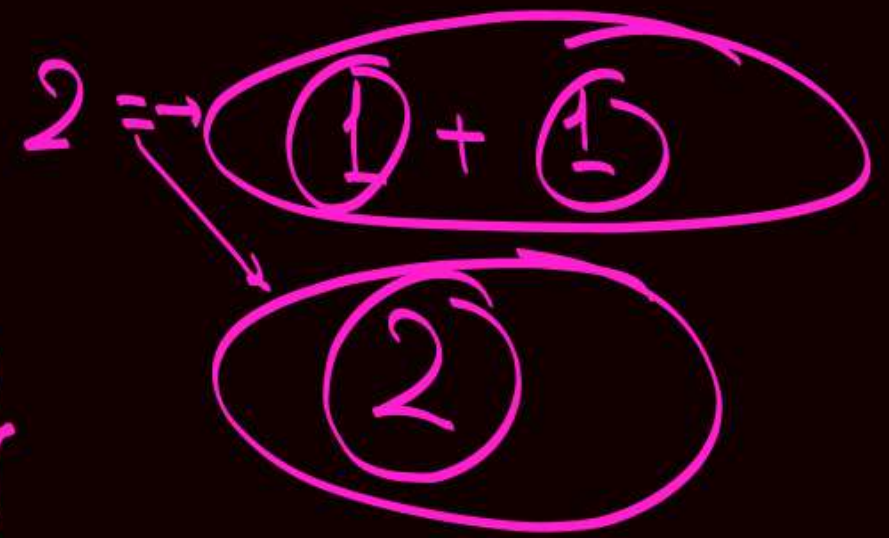
$$\left(\frac{{}^2C_1 * {}^1C_1}{2} \right) = 1 +$$

$|A|=2 \Rightarrow \underbrace{2}$
 it represent
 only one subset
 of size = 2

$\Downarrow$
 $\{\{ _, _ \}\}$

$${}^2C_2 = 1$$

$$\boxed{1 + 1 = 2}$$



④ Let $A = \{a, b, c\}$ i.e., $|A| = 3$

$$|A| = 3 \Rightarrow 1 + 1 + 1$$

$$2 + 1$$

$$3$$

three subsets
of size = 1

one subset of size = 2
and one subset of size = 1

only one
subset of
size = 3.

Partition of set $A = \{\{a\}, \{b\}, \{c\}\}$, $\{\{a, b\}, \{c\}\}$, $\{\{a, b, c\}\}$
 $\{\{a, c\}, \{b\}\}$
 $\{\{b, c\}, \{a\}\}$

i.e., if $|A| = 3$, then number of partitions of set $A = 5$

④ Let $A = \{a, b, c\}$ i.e., $|A| = 3$

$$|A| = 3 \Rightarrow \textcircled{1+1+1}$$

$$|A| = 3 \Rightarrow (1+1+1)$$

$$\text{or } (2+1)$$

$$\text{or } 3$$

$$\textcircled{2+1}$$

$$\textcircled{3}$$

Partition of set $A = \{\{-\}, \{-\}, \{-\}\}$ (or) $\{\{-, -\}, \{-\}\}$ (or) $\{\{-, -, -\}\}$

$$\left(\frac{{}^3C_1 * {}^2C_1 * {}^1C_1}{3!} \right) = 1$$

$$+ \left({}^3C_2 * {}^1C_1 \right) = 3$$

$$+ \left({}^3C_3 \right) = 1$$

$$\text{No. of partitions} = (1 + 3 + 1) = 5$$

$$|A|=1 \Rightarrow 1$$

$$|A|=2 \Rightarrow \overset{(1+1)}{1+1}$$
$$2$$

$$|A|=3 \Rightarrow \overset{(1+2)}{1+1+1}$$
$$1+2$$
$$3$$

$$|A|=4 \Rightarrow \overset{(1+3)}{1+1+1+1}$$
$$1+1+2$$
$$1+3$$
$$2+2$$
$$4$$

$$(1+3)$$

$$|A|=4 \Rightarrow 1+1+1+1$$

$$1+1+2$$

$$1+3$$

$$2+2$$

$$4$$

$$(1+4)$$

$$|A|=5 \Rightarrow 1+1+1+1+1$$

$$1+1+1+2$$

$$1+1+3$$

$$1+2+2$$

$$1+4$$

$$2+3$$

$$5$$

⑤ Let $A = \{a, b, c, d\}$ i.e., $|A| = 4$
How many partitions are there of set A ?

if, $|A| = 4$, then number of partitions of set $A = 15$

Explanation on next slide

(1+3)

$$|A|=4 \Rightarrow \underbrace{1+1+1+1} \Rightarrow \left(\frac{{}^4C_1 * {}^3C_1 * {}^2C_1 * {}^1C_1}{4!} \right) = 1$$

$$\underbrace{1+1+2} \Rightarrow \frac{{}^4C_1 * {}^3C_1 * {}^2C_2}{2!} = 6$$

$$\underbrace{1+3} \Rightarrow {}^4C_1 * {}^3C_3 = 4$$

$$\underbrace{2+2} \Rightarrow \frac{{}^4C_2 * {}^2C_2}{2!} = 3$$

$$\underbrace{4} \Rightarrow {}^4C_4 = 1$$

∴ Total no. of
Partitions = 1 + 6 + 4 + 3 + 1

= 15

Ans

⑥ Let $A = \{a, b, c, d, e\}$ i.e., $|A| = 5$
How many partitions are there of set A ?

if, $|A| = 5$ then number of partitions of set $A = 52$

Explanation on next slide

(1+4)

$|A|=5 \Rightarrow$

$$1+1+1+1+1 = \frac{5C_1 * 4C_1 * 3C_1 * 2C_1 * 1C_1}{5!} = 1$$

$$1+1+1+2 = \frac{5C_1 * 4C_1 * 3C_1 * 2C_2}{3!} = 10$$

$$1+1+3 = \frac{5C_1 * 4C_1 * 3C_3}{2!} = 10$$

$$1+2+2 = 5C_1 * \frac{4C_2 * 2C_2}{2!} = 15$$

$$1+4 = 5C_1 * 4C_4 = 5$$

$$2+3 = 5C_2 * 3C_3 = 10$$

$$5 = 5C_5 = 1$$

Total number of
Partitions = $(1+10+10+15+5+10+1)$

$= 52$

Ans

H.W.

⑥

Let $A = \{a, b, c, d, e, f\}$ i.e., $|A| = 6$

How many partitions are there of set A ?

if, $|A| = 6$ then number of partitions of set $A = ?$

NOTE

- ★ If we know the Equivalence relation on set A , then we can obtain the Partition of set A w.r.t. given Equivalence relation.

Partition of the set can be obtained by defining a set that contains all distinct Equivalence classes w.r.t. given Equivalence relation

eg:- let $A = \{1, 2, 3, 4, 5, 6\}$

and $R = \{(1,1), (2,2), (3,3), (4,4), (5,5), (6,6), (1,2), (2,1), (1,3), (3,1), (2,3), (3,2), (4,6), (6,4)\}$ is an Equivalence relation on set A .

Find the Partition of set A , w.r.t. Equivalence relation R

Soln.

$$[1] = \{1, 2, 3\}$$

$$[2] = \{2, 1, 3\} = \{1, 2, 3\}$$

$$[3] = \{3, 1, 2\} = \{1, 2, 3\}$$

$$[4] = \{4, 6\}$$

$$[5] = \{5\}$$

$$[6] = \{6, 4\} = \{4, 6\}$$

same

same

3 distinct
Equivalence
Classes.

Partition of set A w.r.t.
Equivalence relation R is
 $\{\{1, 2, 3\}, \{4, 6\}, \{5\}\}$

eg: Let $A = \{1, 2, 3, 4, 5\}$

and R is an Equivalence relation on set A .

$$R = \{(1,1), (2,2), (3,3), (4,4), (5,5), (1,2), (2,1), (4,5), (5,4)\}$$

We know partition of set A
w.r.t. Equivalence relation R is

$$= \{ \{1, 2\}, \{3\}, \{4, 5\} \}$$

$$\begin{array}{ccc} \Downarrow & \Downarrow & \Downarrow \\ \{1,2\} \times \{1,2\} & \cup & \{3\} \times \{3\} \cup \{4,5\} \times \{4,5\} \end{array}$$

$$= \{(1,1), (1,2), (2,1), (2,2)\} \cup \{(3,3)\} \cup \{(4,4), (4,5), (5,4), (5,5)\}$$

it is same
as relation R

$$= \{(1,1), (2,2), (3,3), (4,4), (5,5), (1,2), (2,1), (4,5), (5,4)\}$$

eg:- let $A = \{1, 2, 3, 4, 5, 6\}$

and $R = \{(1,1), (2,2), (3,3), (4,4), (5,5), (6,6), (1,2), (2,1), (1,3), (3,1), (2,3), (3,2), (4,6), (6,4)\}$ is an Equivalence relation on set A .

We know,

Partition of set A

W.r.t. relation R is $= \left\{ \{1, 2, 3\}, \{4, 6\}, \{5\} \right\}$

$\{1, 2, 3\} \times \{1, 2, 3\}$

$\cup$

$\{4, 6\} \times \{4, 6\}$

$\cup$

$\{5\} \times \{5\}$

$\{(1,1), (1,2), (1,3), (2,1), (2,2), (2,3), (3,1), (3,2), (3,3)\}$

$\cup$

$\{(4,4), (4,6), (6,4), (6,6)\}$

$\cup$

$\{(5,5)\}$

Result will be same as R

NOTE

- ★ If we know the partition of set A , then we can identify the Equivalence relation on set A corresponding to which the given partition of set A is created.

If we perform the self-cross-product of all the subsets of set A present in the partition, and if we union the result of all those self-cross products, then we get the desired Equivalence relation.

NOTE

For every partition of set A , there is unique Equivalence relation, and with respect to Every Equivalence relation on set A there is unique partition of set A .

i.e., there is a one-to-one Correspondance between the set of all Equivalence relations on set A and the set of all partitions of set A .

i.e.

$$\text{Number of Equivalence Relation on set } A = \text{Number of Partitions of set } A.$$



Topic : Number of equivalence relation on a set

Number of Equivalence relation on set A = Number of Partitions of set A.

→ Find the number of Equivalence relations on "set A" when

(1) $|A| = 0$, then number of Equivalence relation on set A = 1

(2) $|A| = 1$, then number of Equivalence relation on set A = 1

(3) $|A| = 2$, then number of Equivalence relation on set A = 2

(4) $|A| = 3$, then number of Equivalence relation on set A = 5

(5) $|A| = 4$, then number of Equivalence relation on set A = 15

(6) $|A| = 5$, then number of Equivalence relation on set A = 52



Topic : Bell Number



Read about Bell number



2 mins Summary



Topic

Equivalence class ✓

Topic

Partition of a set ✓

Topic

Number of equivalence relation ✓

THANK - YOU